

Private, fair tickets on Midnight

Midnight Buildathon · wave 3 · live on Preprod

PROBLEM

Ticketing is broken twice

Scalping

Bots buy in bulk and resell at 3–10×. Organizers and fans lose; the resale rules can't be enforced.

Data exposure

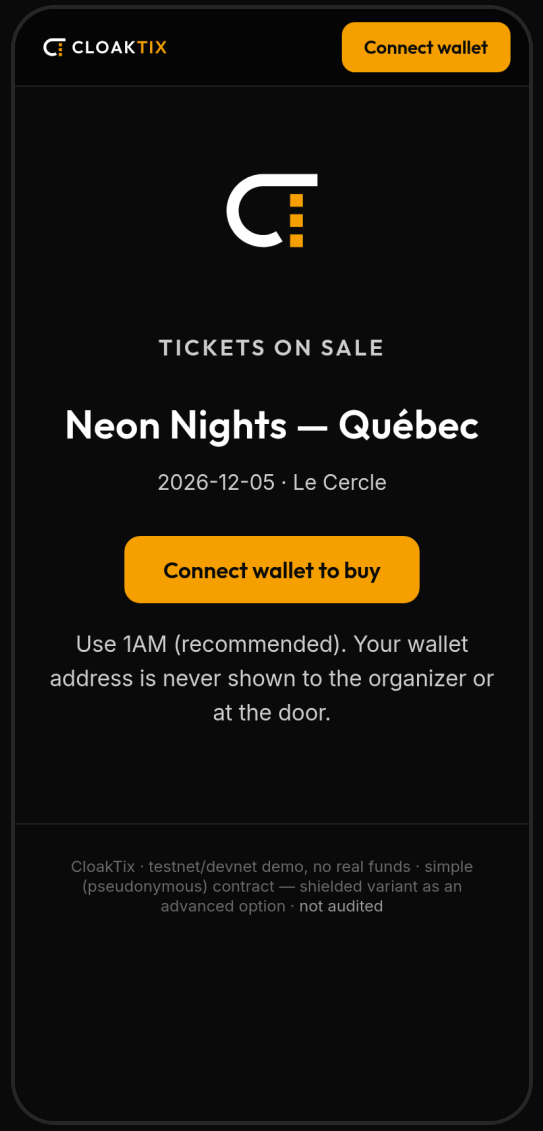
To get in, fans hand over their name, email and sometimes an ID document. Each ticket becomes a data leak.

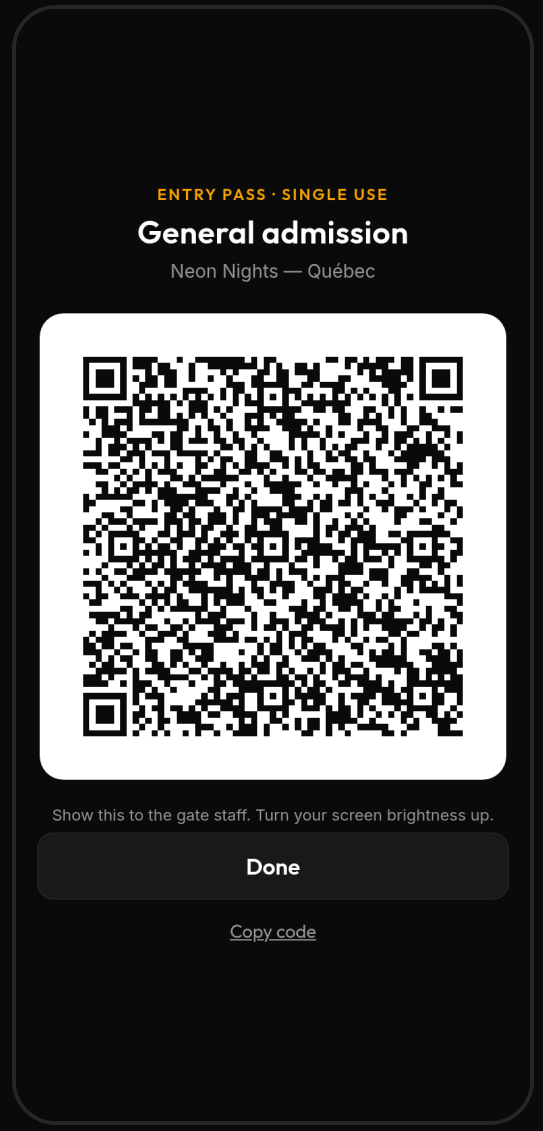
CloakTix: **rules enforced on-chain**, **identity proven in zero knowledge**.

FAN FLOW

Open a link, buy, show a QR

- The organizer shares an event link; it opens on any phone
- Buy with 1AM or Lace. The ZK proof is generated in the browser
- At the door: "Show entry QR" burns the ticket on-chain and returns a **single-use pass**
- The gate admits it once and refuses a replay





ORGANIZER CONTROLS

A resale policy for each ticket type

Capped

Default. Resale price ≤ cap, checked on-chain. Above the cap the listing is refused.

No resale

Listing fails. Check-in still works.

Unlimited

Open market, still with the 3% fee.

18+

A ZK age proof is required. The contract learns only "adult: yes".

No transfer circuit is exported, so the policy can't be bypassed. Age verification will use Midnight Passport once it launches. Demo uses a test issuer.

BUSINESS MODEL

3%

on every sale and every resale

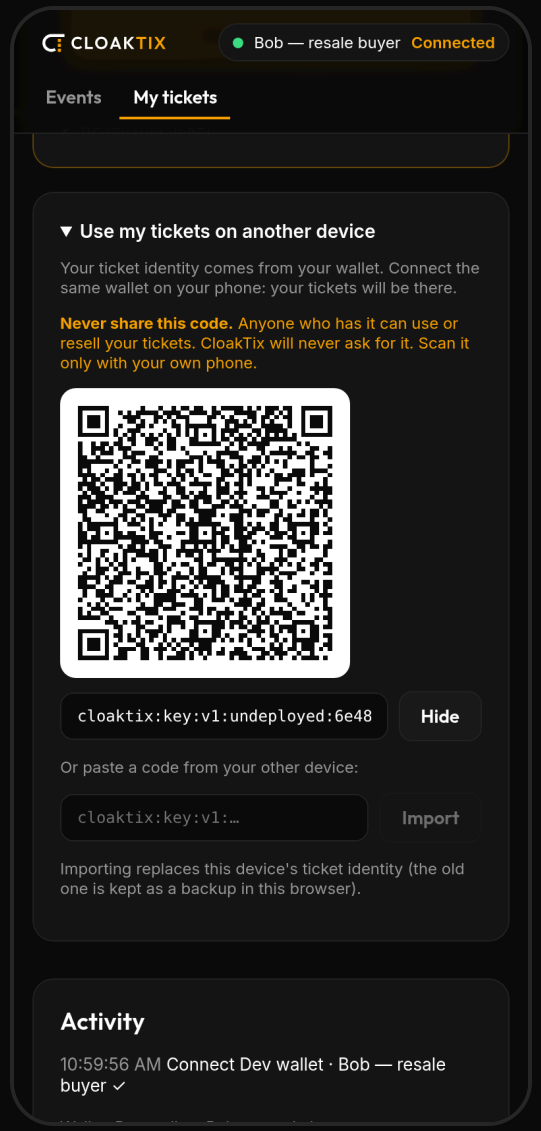
10 tNIGHT → **9.70** organizer · **0.30** fee address

Fee rate and fee address are **sealed** at deploy: immutable, nobody can change them later.

ONE WALLET, EVERY DEVICE

Buy on PC, show it on your phone

- Identity `H(sk)`, with `sk` derived from a domain-separated wallet signature (if deterministic)
- Fallback: a **transfer code** (QR) with a "never share" warning
- Tickets are read from the chain, not from local storage
- Existing users keep their key



ARCHITECTURE

Built on Midnight

Contract

Compact 0.31.1 · 12 circuits · OpenZeppelin MultiToken + Pausable · escrowed resale · check-in nullifiers

Web app

React · Midnight.js 4.1.1 · DApp Connector v4 (1AM first) · in-browser WASM proofs · mobile first

Age

VC-style credential (Jubjub Schnorr) bound to the holder. Test issuer now, Midnight Passport later

Preprod contract

`340b8a07fc5f969c977cee2c41e6386258b46f6f836514a6c154dfe506bcea84`

STATUS & NEXT

Working on Preprod today

Done

Full flow e2e on Preprod. Real 1AM purchase (~45 s proof). Share links, mobile ticket, cross-device identity.

Known limits

~3 wallet approvals per tx · tickets not in 1AM's NFTs tab · NIGHT payments pseudonymous, not shielded · not audited

Next

Midnight Passport issuer · shielded payments · audit · multi-gate sync

Apache-2.0 · built with OpenZeppelin Compact Contracts · cloaktix.com · @CloakTix